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IoT based Wireless Energy Efficient Smart Metering System Using ZigBee in Smart Cities

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***Abstract*—Electricity has become the primary need of human life. The emerging of IoT concept recently in our lives, has offered the chance to establish energy efficient smart devices, systems and cities. Due to the urging need for conserving energy, this paper proposes an IoT based wireless energy efficient smart metering systems for smart cities. A network of smart meters is achieved to deliver the energy consumption data to the Energy/Utility provider. The star and mesh topologies are used in creating the network of smart meters in order to increase the distance of coverage. The proposed system offers an easily operated application for users as well as a Website and database for electricity Supplier Company. The proposed system design has an accuracy level of 95% and it is about 35% lower cost than its peer in the global market. The proposed design reduced the power consumption by 25%.**

***Keywords*—ZigBee, IoT, Smart Metering System, Smart Cities;**

I. INTRODUCTION

The drastic increase in urbanization over the past few years requires sustainable, efficient, and smart solutions for transportation, governance, environment, quality of life, and so on. Since the evolution of Internet of Things (IoT) concept in the early 2000s, the IoTs offers many sophisticated and ubiquitous applications for smart cities. The energy demand of IoT applications is increased. Many researches were presented for smart metering systems and smart grid to reduce the energy consumption by using wireless sensor networks and IoT applications. Various wireless technologies are used in the smart metering application such as GPRS/GSM, WiMAX, Bluetooth, ZigBee, CAN bus and PLC etc. [1] - [4]. The RFID based smart meter was designed but it needs human effort to collect the data for billing [5]. However, the accuracy, reliability of the wireless network, network management, network scalability and the low power operation could be improved in better and cheaper ways.

The conventional energy meters have a great contribution in dissipation of energy. They are post-paid meters, so customers cannot keep track of their energy consumption, except on a monthly basis. Smart meters are electronic measurement devices used by utilities to communicate information for billing customers and operating their electric systems [6]. In this paper, the ZigBee technology is used because of its efficient communication, self-healing networks, low power consumption and scalability. Moreover, ZigBee implementation requires no special permissions since it uses unlicensed 2.4GHz ISM band that is available worldwide. ZigBee has good range, and it works well with networks such as Wi-Fi, Ethernet and GPRS [7] - [9]. The proposed idea is illustrated in Figure 1. As per the illustration, consider a residential area consisting of private houses and each house is installed with the smart meter.

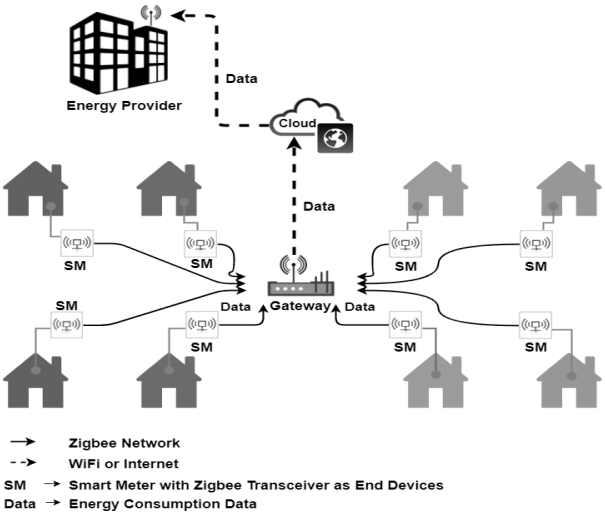


Figure 1: Smart Metering using ZigBee Star Topology.

The smart meter is connected to AC mains supply of the house in order to measure the current and voltage readings. Then the power and energy consumption is calculated. The smart meter has a ZigBee transmitter which sends energy consumption data periodically to the Gateway over ZigBee network. The gateway receives the data and uploads it to the utility provider's cloud/database using internet connection. The utility provider monitors the data and manages the customer's billing information.

The paper is organized as follows: section II gives a detailed explanation of the proposed system. Implementation of ZigBee Network is presented in section III. In section IV, the calibration procedure and the proposed system flowchart are investigated. The Experimental Results of Smart Energy Meter are discussed in section V. Finally, the conclusion is given in section VI.

II. PROPOSED SMART METERING SYSTEM DESIGN

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The proposed system consists of Current Sensor, Voltage transformer, Micro-controllers, display unit, Memory unit ZigBee TX and ZigBee RX as shown in Figure 2.

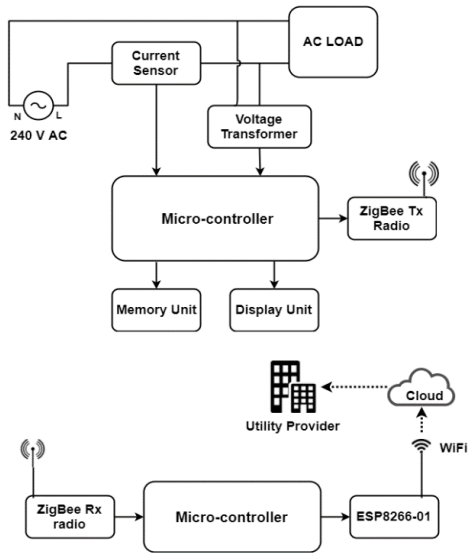


Figure 2: Block Diagram of Smart Energy Meter System.

1. Current Sensor Module:

ACS712-30A, which is Hall effect-based linear current sensing module with the measuring range of -30A to +30A is used as current sensor module. The applied current flowing through it generates the magnetic field which the Hall IC converts into a proportional voltage at the output. This module has the output sensitivity of 66mV per Amp [10].

The current sensor module connected in series with the line of AC mains outputs analog voltage ranging from 0 to 5v. The output of the current sensor is connected to an analog input of the microcontroller with a 10-bit analog to digital converter. This means it will map the input voltage ranging from 0 to 5v to integer values from 0 to 2^{10}-1(1023). This produces a resolution between readings of 5 volts / 1024 units or 0.0049 volts (4.9 mV) per unit.

The AC signal is sampled for one second which is almost 10,000 times in order to calculate accurate maximum and minimum peak values. Now the sampling of AC signal gives us the V_{PP} (peak-peak voltage), from which peak and RMS Current value can be calculated using the below equations:

V_{PP} = ((maxValue - minValue) * 5.0)/1024; -------- (1)
V_{PEAK} = V_{PP}/2; -------- (2)
V_{RMS} = (V_{PEAK}) * 0.707; -------- (3)
I_{RMS} = ((V_{RMS} * 1000) / 66); -------- (4)
where scale factor/voltage sensitivity is 66 mV per Amp.

2. Voltage Transformer Module:

ZMPT101B is the voltage transformer used to measure the AC voltage up to 250 V. It has high accuracy and consistency in measurement of voltage and power [11]. It comes with the inbuilt multi-turn trim potentiometer to adjust output ADC value. The user needs to calibrate using a trim potentiometer to get required ADC output. After sampling the AC signal for one second, maximum and minimum values obtained. Then equation (1), (2) and (3) are used to calculate the RMS voltage.

3. ZigBee Module

In this design, ZigBee S2 series radios are being used which operates on 2.4GHz frequency for transmission and reception of data. They are configured in API mode to send data in frames. The ZigBee radio at the transmitting end is configured as end device with sleep mode enabled so that it sleeps for the period of inactivity and hence saving the power. The software called Digi XCTU is used to configure the ZigBee radios in different modes. The transmission of data from each of the smart meter is scheduled to avoid overlapping of frames at the receiver.

At the gateway, the ZigBee Rx radio receives the data frame sent by the ZigBee Tx radio at transmitting end (smart meter). The received data frame is serially transmitted to the micro-controller using Tx and Rx pins. The microcontroller (Arduino Nano) extracts the RF data and the corresponding device address from the received frame

4. Control Unit:

The control unit consists of two micro-controllers; Arduino Mega 2560 R3 micro-controller is used in the smart meter or transmitting side and Arduino Nano v3 micro-controller is used on the gateway or receiver side. Data received from the sensing unit are passed as an input to the built-in analog to digital converter (ADC) in the microcontroller. Afterwards, the data is processed then used units in kilowatt hour and remaining balance and units.

A 0.96-inch OLED display is used in this implementation to display the energy consumption details to the user. An SD card module is also being used to store the energy consumption data onto an SD card for the sake of backup in case of failure in the communication of data to the receiver.

Afterwards, the control unit issues warnings for the user when either 20% or 10% is left in the balance. At zero balance, 24-hours allowance time is given to user, as a chance to recharge before electricity is cutoff. The calculated values are then passed to the Management unit.

5. Management Unit:

Now the extracted energy consumption data and the corresponding device address is uploaded to the utility provider's cloud using ESP8266-01 Wi-Fi module which is configured to operate in station mode (client). The ESP8266-01 Wi-Fi Module is a SOC with integrated TCP/IP protocol stack which is configured using the AT commands through serial communication [12] [13]. There should be internet or Wi-Fi connection available to the ESP8266-01 module to connect to the web and upload data to the cloud. The ESP8266 module is sent to the sleep mode after it successfully uploads all the available data to cloud conserving the power. The utility provider accesses this data from the cloud and issues the electricity bill to the customers. The database of the users' information and the website are developed. The database is created using XAMPP MYSQL database, used to store

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different billing data of each customer such as first name, last name, mobile number, meter ID, balance, remaining balance used units and remaining units. The website is designed using NetBeans IDE software, HTML and Java servlet.

This unit's functionality is allowing the utility to have control over the customers' billing data and to have a record of this data to track the electricity consumption. The website enables the utility to add new customer, delete, update or view an existing customer's record. It can be accessed by the users using mobile application.

The Android application was implemented using MIT App Inventor 2 Software. Its main functionality is to establish communication between the customers and the managing unit. Additionally, it enables the user to track his/her usage at any time and view metering data on the Android application. The reason behind integrating an android application with the prepaid smart energy meter system is that it will make it easier for customers to monitor their energy consumption even if they are not near the meter display screen.

III. IMPLEMENTATION OF ZIGBEE NETWORK:

The ZigBee network is implemented in two network topologies namely star and mesh networks [14]. The implementation of star network in the smart metering application is easier when compared to that of the mesh topology. Because star network consists of the only parent node (gateway) and child node (smart meter), whereas mesh network comprises of router node along with the parent and child node. The router node in mesh topology routes the data from child node to parent node when the child node is out of coverage area. The mesh topology increases the coverage distance in the smart metering application.

The parent node creates the network and monitors it. Each network will have one parent node and its unique channel and PAN ID. Every node with the same PAN ID belongs to the same network and communicates with each other. Every node will have the destination address to which the data is to be transmitted. Therefore, the child node and the router node will have its destination address as the address of parent node. The parent node will have its destination address as "00" which refers to all the nodes in the network.

The star topology implemented in the smart metering application is illustrated in Figure 1. Similarly, the mesh topology is also implemented with an extra router node between the child and parent node.

IV. CALIBRATION RESULTS

Before designing the final PCB layout, a prototype PCB was designed to check for any instability or errors in the circuit, such as components or connection error. The meter's calibration was done by turning on two refrigerators for five hours and the reading of the already installed meter was taken, the difference between the initial and final readings was 0.5kWh. At the same time, the reading from the implemented circuit design was recorded, the reading differed by 0.01 kWh, the meter read a value of 0.51kWh, when clipped over a single current wire of the two refrigerators connected through the same plug.

Figure 3 illustrates the final system working mechanism. The system operation starts by taking inputs from the current and voltage transformer. Then these inputs are passed to the analog to digital converter of the microcontroller unit where the billing data are calculated. Then a check is done on the billing data. The first check is whether the balance reached 10% or 20%. If the balance reached these low percentages warnings are issued. The second check is whether the balance is finished which if true the electricity is cut off and then the data is transmitted through ZigBee Module and gateway to the database in the cloud that is managed through the website by the utility. The data will be available to the customers by the android application.

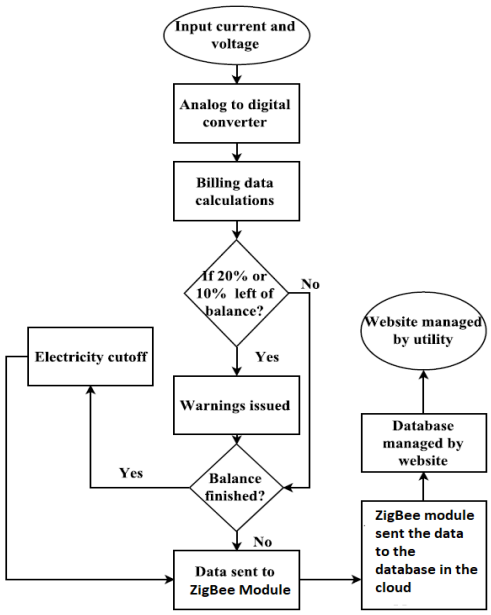


Figure 3 Proposed System Flowchart

V. EXPERIMENTAL RESULTS OF SMART ENERGY METER:

The proposed concept is tested in the test environment with test cases. The coverage distance of 350 meters between the parent node and child node is achieved in case of start network topology with both the transmitter and receiver kept 6 feet above the ground level in the residential area. The coverage distance will increase if the ZigBee radios are installed at greater heights from the ground level. In case of mesh network topology, there will be a router node which also functions as the regular smart meter will be in between the non-reachable child node and parent node. Since router node passes the child's data to parent, the coverage distance is further increased to >500 meters. In a residential area, each house or a building will be at a range of 50 to 100-meter separation. Therefore, this implementation will work at its best and is suitable for the smart metering application.

The smart meter works on 5v DC and 3.3v DC for the majority of its internal components and consumes less than 1 ampere of current. ZigBee radios in sleep mode consume current in terms of microamperes and in active working mode; they consume in milliamperes contributing to the low power

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operation. With sleep modes enabled in ZigBee radios, power consumption reduces by 49% and for ESp8266-01 Wi-Fi module at the receiver end, the power consumption reduces by 87%. On the whole, 71% power reduction is achieved with sleep modes with proposed design.

The sensing units, current sensor module and voltage transformer module have excellent accuracy and consistency in measuring the current and voltage readings. The proper calibration of above modules plays significant role in accuracy. The household appliances like the vacuum cleaner, fan etc. which consumes more power in short time are tested with the smart meter prototyped. The vacuum cleaner with power label 2KW is tested and obtained 2.1KW from the meter; also fan with the label of 50W is tested. Almost 95% accuracy is achieved when compared with the power consumption labels on the electrical appliances. The power consumption of the smart meter designed is 0.75WH which is 25% less than the other meters in the market which consume 1WH or more. The cost of the smart meter developed in this implementation is about 45 Euros which is 35% cheaper when compared with the one in today's market.

The comparison between the features, efficiency and cost of the proposed system and the research paper published in India [15] are presented in the Table 1. The proposed system has good accuracy in measurements, low cost, low power devices, reliable networking capabilities and user-friendly displays and websites.

Table 1: Comparison between proposed meter and other meter.

Feature	Proposed Design	Research Paper Design [15]
Communication Type	ZigBee	ZigBee
Network Topology	star and mesh	Only star
Payment Method	Postpaid (No hassle of power cuts due to non-payment). Fine when not paid.	Prepaid and postpaid.
Back-end interface	Website and database	ZigBee Receiver
Data Retransmissions and ACKs	Yes in case of communication failure.	No
Data backup	SD card in smart meter.	No
Low power Operation	Yes, Sleep modes enabled.	Yes, Not used sleep modes.
Power Consumption by Meter	0.75WH	1WH
Power Reduction Achieved	25%	NA
Cost	45 Euros	~70 Euros

VI. CONCLUSION:

The conservation of energy is much needed in the modern world. This proposed design of smart metering contributes to energy conservation and also reduces human effort in billing. The user just needs to buy and install the smart meter. The user needs not to have an internet connection or any smart devices from his end for sending data to the utility provider. But internet should be available at each gateway point which is provided by utility provider according to this design. The application of mesh networking increases the coverage area. Otherwise, star network would be sufficient.

The proposed smart meter design is user-friendly, which makes the user aware of their energy consumption by displaying the details. Customers can also track their energy consumption from the web and mobile apps from anywhere in the world, with restricted access and user authentication. The proposed design has 95% accuracy, 25% less power consumption and 35% cheaper when compared to existing energy meters in the market. As per this model, not only the smart meters are of low cost, but the entire infrastructure can be built at low cost.

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